ACTIVITY 3: Enhancing the Account Class

Objective

Add validation, business rules, and security features to the `Account` entity class using boolean returns.

Requirements

Enhancement 1: Age Validation

- Account holders must be at least 18 years old
- If age < 18, constructor sets age to 18 automatically

Enhancement 2: Account Type Validation

- Only "Savings" and "Current" are valid
- If invalid, automatically default to "Savings"

Enhancement 3: Minimum Balance Rules on Creation

- Savings accounts: Minimum balance ₹500
- Current accounts: Minimum balance ₹1000
- If initial balance is below minimum, automatically set to minimum

Enhancement 4: Minimum Balance Enforcement on Withdrawal

- After withdrawal, balance must not fall below minimum
- If it would, return `false` from `withdraw()`

Enhancement 5: Account Status Management

- Add `closeAccount()` method: Sets status to "Inactive", returns `boolean`
- Add `reopenAccount()` method: Sets status to "Active", returns `boolean`
- `deposit()` and `withdraw()` return `false` if account is inactive

Enhancement 6: PIN Protection

- Add `pin` field (Integer, 4-digit, can be null)
- Add `setPin(int pin)` method: Returns `boolean` (true if valid)
- Add `verifyPin(int pin)` method: Returns `boolean`
- Add `hasPin()` method: Returns `boolean`
- Modify `withdraw()` to accept PIN parameter and return `boolean`

Modified Methods

Constructor

```
public Account(int accountNumber, String name, int age,
        double initialBalance, String accountType)
```

- If age < 18, set to 18
- If account type invalid, default to "Savings"
- If balance below minimum, set to minimum

withdraw (with PIN)

```
public boolean withdraw(double amount, int pin)
```

- Check account is active
- Verify PIN
- Check minimum balance after withdrawal
- Return `true` if successful, `false` otherwise

deposit (with status check)

```
public boolean deposit(double amount)
```

- Check account is active
- Return `false` if inactive
- Return `true` if successful

New Methods

Method	Parameters	Return Type	Description
`closeAccount`	None	`boolean`	Returns `true` if closed, `false` if already closed
`reopenAccount`	None	`boolean`	Returns `true` if reopened, `false` if already active
`setPin`	`pin` (int)	`boolean`	Returns `true` if valid (4-digit), `false` otherwise
`verifyPin`	`pin` (int)	`boolean`	Returns `true` if matches, `false` otherwise
`hasPin`	None	`boolean`	Returns `true` if PIN is set

Important Notes

- **NO display methods** - pure entity class
- Constructor self-corrects invalid values

Task

Create an enhanced `Account` class with all the above features in a file called `AccountEnhanced.java`.